ingredients in string theory could similarly be molded together to yield particular black holes.

In actuality, black holes are one possible end product of stellar evolution. After a star has burned all its nuclear fuel through billions of years of atomic fusion, it no longer has the strength—the outward-directed pressure—to withstand the enormous inward force of gravity. Under a broad spectrum of conditions, this results in a cataclysmic implosion of the star's enormous mass; it violently collapses under its own tremendous weight, forming a black hole. Contrary to this realistic means of formation, Strominger and Vafa advocated "designer" black holes. They turned the tables on black hole formation by showing how they could be systematically constructed—in a theorist's imagination—by carefully, slowly, and meticulously weaving together a precise combination of the branes that had emerged from the second superstring revolution.

The power of this approach became immediately clear. By maintaining full theoretical control over the microscopic construction of their black holes, Strominger and Vafa could easily and directly count the number of rearrangements of the black hole's microscopic constituents that would leave its overall observable properties, its mass and force charges, unchanged. They could then compare this number with the area of the black hole's horizon—the entropy predicted by Bekenstein and Hawking. When Strominger and Vafa did so, they found perfect agreement. At least for the class of extremal black holes, they had succeeded in using string theory to account for the microscopic constituents and the associated entropy precisely. A quarter-century-old puzzle had been solved.6

Many string theorists view this success as an important and convincing piece of evidence in support of the theory. Our understanding of string theory is still too coarse to be able to make direct and precise contact with experimental observations of, say, the mass of a quark or an electron. But we now see that string theory has provided the first fundamental explanation of a long-established property of black holes that has stumped physicists using more conventional theories for many years. And this property of black holes is intimately tied up with Hawking's prediction that they should radiate, a prediction that, in principle, should be experimentally